

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - X

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PART B – Questions

- a. **Create a PWA with a Web App Manifest file including: App name Icons Start URL Display mode Implement Push Notifications in a PWA.**
- b. **Develop an E-Commerce PWA with product listing and offline cart functionality.**

Program.cs

```
var builder = WebApplication.CreateBuilder(args);  
var app = builder.Build();  
app.UseDefaultFiles();  
app.UseStaticFiles();  
app.Run();
```

manifest.json

```
{  
  "name": "E-Commerce PWA",  
  "short_name": "ShopPWA",  
  "start_url": "/index.html",  
  "display": "standalone",  
  "background_color": "#ffffff",  
  "theme_color": "#d6d6d6",  
  "icons": [  
    {  
      "src": "icons/icon1.png",  
      "sizes": "192x192",  
      "type": "image/png"  
    },  
    {  
      "src": "icons/icon2.png",  
      "sizes": "512x512",  
      "type": "image/png"  
    }  
  ]  
}
```

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>E-Commerce PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#d6d6ed">
  <style>
    body {
      font-family: Arial;
      margin: 20px;
    }
    .container {
      display: flex;
      gap: 30px;
    }
    .cart-section {
      width: 30%;
      border: 1px solid gray;
      padding: 15px;
      min-height: 300px;
    }
    .product-section {
      width: 70%;
    }
    .products {
      display: flex;
      flex-wrap: wrap;
      gap: 15px;
    }
    .product {
      border: 1px solid gray;
      padding: 10px;
      width: 200px;
    }
    button {
      padding: 5px 10px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h1>E-Commerce PWA</h1>
  <div class="container">
    <!-- Cart Section -->
    <div class="cart-section">
      <h2>Cart Items</h2>
      <ul id="cartList"></ul>
    </div>
```

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```
<!-- Product Section -->
<div class="product-section">
  <h2>Products</h2>
  <div class="products">
    <div class="product">
      <h3>Laptop</h3>
      <p>Price: ₹ 50000</p>
      <button onclick="addToCart('Laptop')">
        Add to Cart
      </button>
    </div>
    <div class="product">
      <h3>Mobile</h3>
      <p>Price: ₹ 20000</p>
      <button onclick="addToCart('Mobile')">
        Add to Cart
      </button>
    </div>
    <div class="product">
      <h3>Headphones</h3>
      <p>Price: ₹ 3000</p>
      <button onclick="addToCart('Headphones')">
        Add to Cart
      </button>
    </div>
  </div>
</div>
</div>
</div>
<script src="app.js"></script>
</body>
</html>
```

app.js

```
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('/service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}
let cart = JSON.parse(localStorage.getItem('cart')) || [];
function addToCart(product) {
  cart.push(product);
  localStorage.setItem('cart', JSON.stringify(cart));
}

function displayCart() {
  let cartList = document.getElementById('cartList');
  cartList.innerHTML = "";
  cart.forEach(item => {
    let li = document.createElement('li');
    li.textContent = item;
    cartList.appendChild(li);
  });
}
```

```
});  
}  
service-worker.js  
const CACHE_NAME = 'ecommerce-cache-v1';  
const urlsToCache = [  
  '/',  
  '/index.html',  
  '/app.js',  
  '/manifest.json'  
];  
self.addEventListener('install', event => {  
  event.waitUntil(  
    caches.open(CACHE_NAME)  
      .then(cache => {  
        return cache.addAll(urlsToCache);  
      })  
  );  
});  
self.addEventListener('fetch', event => {  
  event.respondWith(  
    caches.match(event.request)  
      .then(response => {  
        return response || fetch(event.request);  
      })  
  );  
});
```

